AI-Driven RF Design: An LLM Coding Agent Takes a Substrate Integrated Waveguide Filter from Specification to Fabrication

Federico Borello
Universidad Tecnológica Nacional — Facultad Regional Buenos Aires, Argentina
fborello.contact@gmail.com

***Abstract*—We report the design, fabrication and measurement of a third-order, 2.87 GHz substrate integrated waveguide bandpass filter in which a large-language-model coding agent wrote and operated the entire toolchain: Chebyshev synthesis, parametric geometry, finite-element simulation with the open-source solver Palace, Bayesian optimization, and the export of fabrication files. One engineer set the specification, took the design decisions and ran the commercial-solver cross-checks. Three results stand out. First, a Gaussian-process optimizer applied to the assembled filter never reached a usable match, because the pass-band width it scores is discontinuous in the external coupling; measuring smooth observables on small structures and inverting the coupling relations converged in about three solves per curve. Second, most of the 47 fixes in the project history corrected defects that produced plausible numbers without raising any error; we classify them and list the checks that now catch each class. Third, the board, made by JLCPCB on Rogers RO4350B with ENIG finish for about USD 120, was measured with a vector network analyzer and recalibrated independently in software. It is centred at 2.894 GHz with 171 MHz bandwidth, 2.0 to 2.4 dB insertion loss and one broad reflection minimum near -18 dB, the floor of the calibration check. The simulation predicted 2.876 GHz, 182 MHz, 1.7 dB and three zeros, and the measured loss fits an effective unloaded Q of about 130 to 185. The -16 dB specification is not met.**

***Index Terms*—Substrate integrated waveguide, bandpass filter, large language models, AI agents, design automation, Bayesian optimization, finite-element method, coupling matrix.**

Introduction

Machine learning has been part of microwave CAD for decades. Neural-network surrogate models [1], space mapping [2] and Bayesian optimization [3], [4] all replace some of the expensive electromagnetic (EM) solves with a cheaper learned or coarse model. Learned models are also used to extract coupling matrices from simulated responses, recently with physics-informed networks that need no labelled training set [5]. In each of these approaches the engineer still writes the geometry scripts, meshing, port setup, objective and export code, and interprets what the solver returns.

Recent large-language-model (LLM) agents work differently: they write and run code in a loop, observe the results and revise [6], [7]. Agent frameworks now exist for RFIC design [8] and RF amplifiers [9]. An agent has also operated CST, ADS and KiCad through their scripting interfaces to design a GNSS front end [10]. Empirical studies of bugs in agentic frameworks classify failures of the agent software itself [11]. What these reports leave largely unmeasured is how the *engineering* fails when an agent drives it: defects in the physical model that the agent writes, which matter more than the successes when one decides how much to trust the numbers.

This paper reports one complete design effort of that kind. The object is a classic structure, a three-cavity Chebyshev SIW filter [12], [13], [14]. It was chosen because its physics is textbook, so every agent-produced number can be checked against theory. An LLM coding agent (Claude Code [15]) wrote and ran all of the design software. The author acted as specifier, reviewer and decision-maker. The project history is preserved in git, and most claims below map to a commit or a stored result. Some process statistics, such as the screen-out rates of the Bayesian search, survive only in contemporaneous engineering notes rather than raw logs; we flag these where they appear.

The contributions are:

1. An account of the tool path: an FDTD solver (openEMS) that could not support match tuning, a finite-element solver (Palace) that could, and a GPU workstation run through remote agent sessions whose memory limits eventually pushed all production runs onto CPUs (Sec. III).
2. A measured explanation of why black-box Gaussian-process (GP) search over the assembled filter fails, and of why calibrate-and-invert succeeds (Sec. IV).
3. A taxonomy of the silent modelling defects the process produced and caught, with the guard that now blocks each class (Sec. V).
4. The fabricated design, with its deviation from specification diagnosed by coupling-matrix extraction. It is accompanied by a documented dual-mode negative result, an insertion-loss budget, a launch study, the manufacturing path and the VNA measurement of

the fabricated board (Secs. VI to VIII).

Filter specification and synthesis

The specification is f0 = 2.87 GHz, 5 % fractional bandwidth (FBW, 143.5 MHz), 0.1 dB Chebyshev ripple and $|S11| \le -16$ dB in band. The laminate is Rogers RO4350B, 0.76 mm thick ($\varepsilon_r = 3.66$, $\tan\delta = 0.0037$ [16]), with 0.5 oz copper. The fence vias are 1.5 mm in diameter at a 3.0 mm pitch. Their p/d = 2 ratio sits at the edge of the leakage rules of [17], [18]. Smaller vias (0.8 mm at 1.5 mm pitch) were compared three times under three different objectives and never matched this choice. The connector is a Taoglas EMPCB.SMAFSTJ.B.HT edge-mount SMA [19].

Design equations

The low-pass Chebyshev prototype gives $g_1 = g_3 = 1.0316$ and $g_2 = 1.1474$ for $N = 3$ and 0.1 dB ripple [20]. The bandpass mapping turns them into the two numbers the geometry must reproduce, the coupling between adjacent cavities and the external quality factor of the end cavities:

$$k_{i,i+1} = \frac{\mathrm{FBW}}{\sqrt{g_i\, g_{i+1}}} = 0.0460, \qquad Q_e = \frac{g_0\, g_1}{\mathrm{FBW}} = 20.63. \tag{1}$$

The waveguide width follows from choosing the TE10 cutoff near $0.7 f_0$, which sets the equivalent width, and from the Cassivi–Wu correction for the finite via fence [21]:

$$a_{\mathrm{eff}} = \frac{c}{2 f_c \sqrt{\varepsilon_r}}, \quad f_c \approx 0.7 f_0, \qquad a_{\mathrm{siw}} = a_{\mathrm{eff}} + \frac{d^2}{0.95\, p}. \tag{2}$$

Each cavity length follows from the TE101 resonance condition:

$$f_{101} = \frac{c}{2\sqrt{\varepsilon_r}} \sqrt{\frac{1}{a_{\mathrm{eff}}^2} + \frac{1}{L^2}}. \tag{3}$$

Eq. (3) assumes ideal walls, so the real length is shorter once the tapers and apertures load the cavity (Sec. VI). The feed is a 50 Ω microstrip line (Hammerstad [22]) flared into the SIW by a Deslandes taper [23]. At the board edge it becomes a grounded coplanar waveguide (CPWG) launch sized to the SMA footprint: a 1.2 mm trace with 0.2512 mm gaps, which solves to 50.0 Ω by conformal mapping [24], [25].

The two couplings are measured, not computed, because closed-form iris formulas are unreliable for SIW. Two coupled closed cavities split their resonance into f_1 and f_2, and a single cavity driven through its apertures has a loaded quality factor Q_L:

$$k = \frac{f_2^2 - f_1^2}{f_2^2 + f_1^2}, \qquad Q_e = \frac{2}{1/Q_L - 1/Q_u}. \tag{4}$$

The coupling-matrix formulation [26], [27] provides the ideal target response and, later, the diagnostic fit. The dissipation of a narrowband filter then follows from the unloaded quality factor Q_u [20]:

$$\mathrm{IL}_0 \approx \frac{4.343 \sum_i g_i}{\mathrm{FBW}\, Q_u}\ \mathrm{dB}. \tag{5}$$

Toolchain and working method

Fig. 1 shows the flow and the division of labour. Fig. 2 shows how the effort was distributed in time.

Phase 1: analytic design, openEMS and CST

The first iteration (June 2026) used analytic dimensions, the free FDTD solver openEMS [28] for calibration, and CST Studio Suite 2021 [29] as the reference. Python generated a parametric CST VBA macro that builds native solids wrapped in the solver's history list. The analytic cavity length of 38.23 mm put the CST passband at 2.818 GHz with a 293 MHz band, 3.65 dB insertion loss and only 1.02 dB return loss. The external coupling was uncalibrated and heavily over-coupled.

openEMS was abandoned for three reasons. First, FDTD resolves the 1.5 mm cylindrical vias on a staircase grid, which needs at least three cells per diameter over the whole board. The resonant, high-Q structure also rings down slowly, so each full-filter run took minutes to tens of minutes. Second, the ports did not give a trustworthy S11. The lumped port radiated like a point source instead of launching the microstrip mode. The microstrip port launched the mode correctly, but its S11 de-embedding was unstable and even returned $|S11| > 1$. The passband and S21 were reproduced, but the match could not be tuned, and matching was the whole remaining problem. Third, there was no eigenmode solver for the coupling calibration of Sec. IV, which had to be emulated by reading peak splits off driven sweeps.

Palace [30], an MPI-parallel finite-element solver built on MFEM [31] with gmsh meshes [32], removed all three obstacles. Its tetrahedral mesh conforms to the vias. Its wave ports solve the 2D port mode, as CST's do, and are passive by construction. It has an eigenmode solver, and it is driven entirely by JSON files, which makes it easy for an agent to operate. A first Palace model reproduced the CST result for the same geometry and then replaced openEMS in the loop.

Phase 2 onward: an agent-operated toolchain

From August the agent consolidated three copied optimizer scripts into one package with a `Topology` interface and a single command-line tool (`siw`). The tool covers synthesis, layout, eigenmode and driven solves, Bayesian design, 1-D scans, calibration, comparison figures and CAD export. Every production number in this paper is reproducible through that tool, and curated results

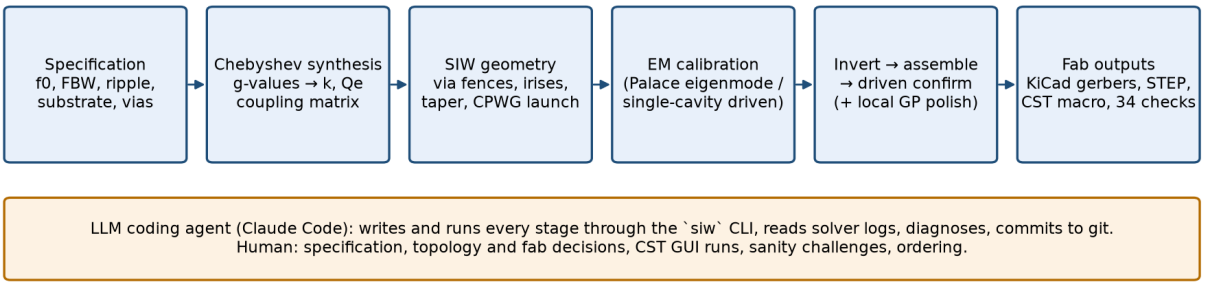


Fig. 1. Design flow. The agent writes and operates every stage through one command-line tool (`siw`); the engineer supplies the specification and the decisions, and runs the commercial-solver cross-checks.

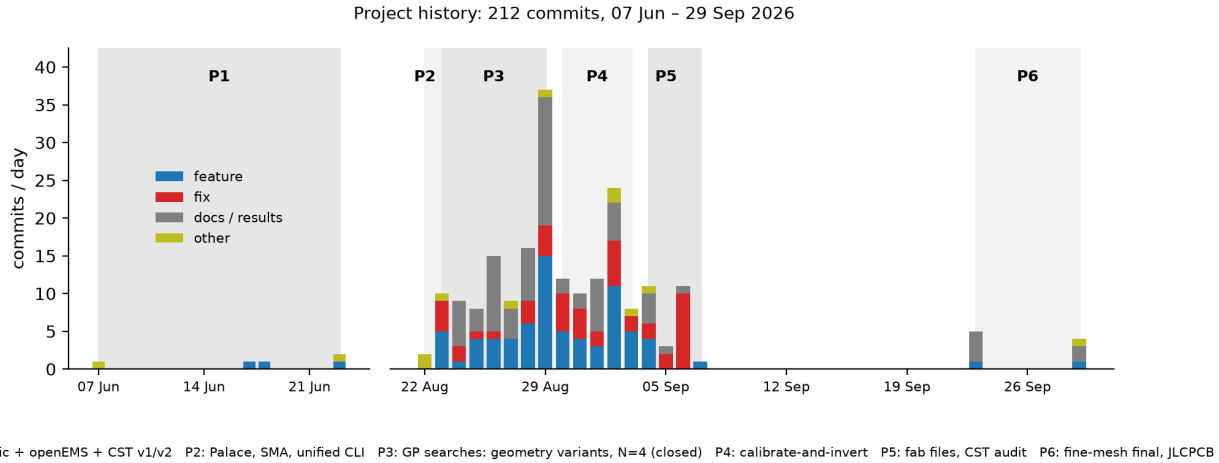


Fig. 2. Commit history by type. P1 was a human-led prototype committed in a few large commits. From P2 onward the agent drove the work, and fixes (red) cluster where new physics or a new tool entered the loop.

are stored in a `golden/` directory together with their S-parameters and the command that reproduces them. A project brief file (`CLAUDE.md`) acted as the agent's persistent memory across sessions. Handoff documents carried the state from one machine to another.

From the textbook steps to the tool. Table I lists what a coupled-resonator SIW filter normally takes by hand and which command replaces each step. The equations are the classical ones [18], [20], [21]. What the tool changes is that every step that used to be a hand-fitted curve is a measurement with a solver, and the chain from specification to gerbers is one sequence of commands.

The optimizer. The search code is our own, written around Palace, not a package taken from a design tool. A candidate is a set of geometry parameters: the cavity lengths (mirrored about the centre), the iris widths, the external aperture and the taper length. The microstrip width stays at its 50 Ω value. To evaluate a candidate the code meshes the whole board with gmsh, connectors and launch included, writes the Palace configuration, runs a driven sweep on a coarse mesh and reads $|S_{11}|$ and $|S_{21}|$. A fast physics screen at f_0 runs first and rejects candidates that do not resonate; the rejected ones get a graded penalty, so the surrogate still sees a gradient. The objective sums the RMSE of $|S_{11}|$ against the ideal Chebyshev response (both clipped at a floor), the shape of $|S_{21}|$, the insertion loss, the centre-frequency error outside a deadband, a two-sided bandwidth error and the board length. Three one-sided hinges add the real specification: worst in-band $S_{11} \leq -16$ dB, loss within an allowance over the dielectric floor, and symmetry. The objective went through ten revisions (v9–v18); the early ones scored individual points and the later ones fitted the whole response. A Gaussian-process surrogate (scikit-optimize [33], [34], `gp_hedge` acquisition) proposes each next candidate after a random seeding phase whose size scales with the budget. A patience rule stops a plateau early, and the winner is solved again on a fine mesh and checked after rounding every dimension to the 0.1 mm fabrication resolution. In local mode (`--seed`, `--span`) the search box is a few millimetres around a calibrated design. A coarse evaluation takes 6–20 min, so sample efficiency matters more than throughput, and a lock serialises the Palace solves.

TABLE I
THE DESIGN STEPS OF AN SIW COUPLED-RESONATOR FILTER: THE TEXTBOOK WAY AND WHAT THE TOOL DOES.

	Step	By hand	In the tool
1	Synthesis	Chebyshev g_i by recursion; $k_{i,i+1} = \mathrm{FBW}/\sqrt{g_i g_{i+1}}$, $Q_e = g_0 g_1/\mathrm{FBW}$; ideal response from the coupling matrix	`synthesis.py`: targets k, Q_e and the ideal `target.s2p`
2	Waveguide width	$a_{\mathrm{eff}} = c/(2f_c\sqrt{\varepsilon_r})$ with $f_c \approx 0.7 f_0$; $a_{\mathrm{siw}} = a_{\mathrm{eff}} + d^2/(0.95p)$; via rules on p/d	`siw_geometry.py`, from the substrate and via presets in `config.py`; builds the via fences and the board
3	Cavity length	$f_{101} = \frac{c}{2\sqrt{\varepsilon_r}}\sqrt{a_{\mathrm{eff}}^{-2} + L^{-2}}$	first guess only; corrected from a driven solve in step 7
4	Feed	50 Ω microstrip width (Hammerstad), taper (Deslandes), coplanar gap by conformal mapping	`siw_geometry.py`, `launch.py`: taper and the CPWG footprint of the SMA
5	Internal coupling	simulate two coupled cavities, read the split $k = (f_2^2 - f_1^2)/(f_2^2 + f_1^2)$, fit $w(k)$	`siw calibrate --what k`: Palace eigenmode of a closed two-cavity model, polynomial fit, inverted at the target k
6	External coupling	one resonator, Q_L from the −3 dB width or the group delay, $Q_e = 2/(1/Q_L - 1/Q_u)$	`siw calibrate --what qe`: one driven cavity with two apertures, fit of $w_{\mathrm{ext}}(Q_e)$
7	Assembly and tuning	simulate the whole filter, change lengths and irises by hand until the reflection zeros are equiripple	`siw synth-design`: joint (k, f_c) refinement, driven centre-frequency trim, separate middle-cavity correction, pole-shape check
8	Final polish	manual sweeps of one parameter at a time	`siw design --seed`: local Bayesian search (below)
9	Layout and CAD	CST model drawn by hand; gerbers from a PCB tool	`siw cst`, `siw kicad` and 34 file-level checks
10	Verification	VNA measurement and calibration	`siw measure`: software recalibration and comparison with the simulations

Compute. To speed up the search we moved to a workstation with an NVIDIA GTX 1660 SUPER (6 GB) and ran Palace with its GPU backend there. The author supervised the long agent sessions remotely through Claude Code's Remote Control. On the dual-mode model the GPU was 1.24× faster than 4 CPU ranks, with S-parameters identical to 10 dB, and disabling MFEM's pooled allocator (`MFEM_MEMORY=cuda`) removed the first out-of-memory failures, which had killed every point of an eigenmode sweep. But the card does not have the memory for the meshes that matter. It finished a driven solve with 174.6 k tetrahedra and ran out at 178.4 k, in the error estimator, and only one rank per card works. The 166 mm cascade exceeds that at any adequate mesh density, so no GPU mesh preset was ever validated for it; the code now caps GPU use at 176 k tetrahedra and falls back to the CPU. The GPU was not tried on the fine-mesh verification, which needs 315 k unknowns and about 5.5 GB on the CPU. All production runs therefore ran on CPUs with 4 MPI ranks. Four attempts on a 3.7 GB host failed, three of them explicitly out of memory; the final runs used a 16 GB host at about 8 s per frequency point. Memory, not arithmetic, was the binding constraint, and a card with more memory, or the half-model and adaptive sweeps of Sec. X, would bring the cascade onto the GPU.

WHY GLOBAL BAYESIAN SEARCH FAILED

The GP worked as intended in one regime: as a local polish around a validated design. For the dual-mode filter, a 30-evaluation search in a ±2 mm, 5-dimensional box confirmed the landed geometry. Neighbouring probes came back worse, and the apparent 0.41 dB gain lay inside the measured coarse-mesh noise floor of 0.64 dB.

Run globally over the assembled cascade, the same machinery never produced a worst in-band S11 better than −5 dB in more than 60 driven evaluations. The cause is structural. The objective scores the passband width, and that width is discontinuous in the external aperture w_{ext}. Below a coupling threshold only the centre pole clears the −3 dB line, so the band measures one resonance. Above the threshold all N poles merge and the width jumps. On the N = 3 cascade the widths recorded during the search were 16, 22, 36, 65 and then 215 MHz, with the 143.5 MHz target inside the jump. We re-measured this for the paper on the fabricated geometry (Fig. 3c, data in the repository). The width rises 96, 106, 123 MHz for w_{ext} = 10–12 mm, jumps to 195 MHz at 13 mm, and then *falls* again as the end cavities are over-coupled and the reflection zeros drop from three to one. The target width is crossed twice. The second crossing, at 16 mm, has one reflection zero and −4 dB of return loss; a bandwidth-driven search, or the scan's own inversion (15.2 mm), lands there. A GP assumes a smooth objective, so it inherits both the jump and the false solution. The search-health data showed the same thing: 57–83 % of cascade candidates were screened out, leaving 2–9 informative points per run, against 34–37 % and about 19 informative points on the dual-mode filter, where the search converged.

The replacement measures each coupling on the smallest structure that exhibits it (Fig. 3):

- **Internal coupling.** A closed two-cavity eigenmode solve splits the cavity mode into f_1 and f_2, and k follows from Eq. (4). It is smooth in w_{iris}.
- **External Q.** One cavity driven symmetrically through two apertures gives the loaded Q, and Q_e follows from Eq. (4). Here $Q_u = 1/\tan\delta = 270.3$, independently matched by the eigenmode Q. It is smooth and monotonic in w_{ext}.

Inverting the fitted curves at the synthesis targets, assembling the filter and making a final driven fc trim converged in about three solves per curve. The re-run calibration for this paper inverts to $w_{\text{iris}} = 13.85$ mm and $w_{\text{ext}} = 13.53$ mm, against 14.42 and 13.96 mm on the fabricated board, whose end cavities were subsequently trimmed. The resulting internal coupling was within 2 % of target. At that checkpoint the N = 4 centre frequency also landed at 2.870 GHz, although the N = 4 design was later abandoned, its best return loss being −6.1 dB (Sec. VI). The GP was kept only as a local polish seeded with the result (`--seed`). The physics that worked is textbook [20], [27]. What the agent contributed was running the calibrations quickly, and diagnosing why the black-box route had failed.

Two further search failures were caused by the parameterization itself rather than by the search. The via fences were first generated by filtering a fixed grid, so the iris width was *quantized*. The whole cascade search box spanned about two distinct geometries, and the GP was tuning a knob that did nothing (Fig. 4). In addition, N ≥ 4 had a single shared iris width and could not express the required k = [0.0415, 0.0329, 0.0415].

Silent defects: a taxonomy

As of commit `b06bf63` (29 Sep 2026) the history holds 212 commits, 117 of them with an agent co-author trailer, and 47 labelled *fix*. Most corrected defects that never raised an error: the solver converged and the plot looked reasonable, but the number was wrong. Table II groups representative cases. Three rules paid for themselves repeatedly. First, if a fix does not change the number, the diagnosis was wrong. Second, after two wrong inferences, stop inferring and run a one-factor ladder. Third, check every result against what physics allows: insertion loss cannot beat a lossless model, the pole count must equal N, and a lossy stack cannot cost 27 dB.

Two cases changed the course of the project. In the first, when a real SMA coax launch replaced the ideal port, the optimized microstrip width of 2.26 mm turned out to be a 41 Ω line. The optimizer had exploited a port that normalizes to whatever it sits on, and the true 50 Ω width is 1.664 mm. The first coax model also used a full-size 0.82 mm pin, which is thicker than the 0.76 mm board, and returned −0.2 dB S11. That was real physics, not a bug (Fig. 5). In the second, an independent audit against CST showed that the −13.37 dB return loss used to choose the cascade over the dual-mode design was a coarse-mesh artefact. The converged value is about −10.7 dB. The decision survived, but the justification had to be rewritten.

Topologies, lengths and the dual-mode negative result

Cavity lengths. The analytic 38.23 mm cavity placed the band at 2.82–2.905 GHz (openEMS/CST). An openEMS sweep to 35.93 mm and 33.64 mm moved it to 2.982 and 3.077 GHz, a slope of about −37 MHz/mm. Palace, which models the loading by the tapers and external apertures that the analytic formula ignores, shortened the cavities to about 34.4 mm. Calibrate-and-invert then gave unequal cavities, 34.24/35.50/34.24 mm. The fine-mesh coupling fit found the end cavities about 17 MHz low, and a −0.3 mm trim (33.94/35.50/33.94 mm) was chosen. A −0.6 mm trim gained only 0.07 dB and cost 0.4 dB of band-edge loss. The measured sensitivity at the end cavities was about 27 MHz/mm, roughly half the 50 MHz/mm rule of thumb.

Board size. The design space was also explored for a smaller board (Figs. 6 and 7). The candidates were a single-cavity dual-mode N = 2 filter (119 mm), the N = 3 cascade (166 mm) and an N = 4 cascade (211 mm). Three other single-cavity shapes were tried: corner-cut, circular-cross and rhombic. None came closer to the specification than the dual-mode cavity (best S11 −6 to −9 dB). A compact N = 4 made from two dual-mode cavities was closed after five mechanisms plus a joint GP search all stopped at two of four poles: offset apertures, near-corner coupling, top-copper slots, discrete posts and quad-mode cavities. A four-cavity single-mode cascade works, but it needs 180–206 mm.

Dual-mode N = 2. A single square SIW cavity, a = 57.16 mm, carries a degenerate TE102/TE201 pair. A floating internal via row splits the pair and couples it [35], [36]. Under the matched CPWG launch its best response is fc 2.904 GHz, 316.6 MHz bandwidth, −10.76 dB return loss and 1.45 dB insertion loss. Real copper adds 0.32 dB. One-dimensional scans on the assembled filter show that, on this topology, bandwidth and match are *opposed* on every geometry parameter. At $w_{\text{iris}} = 11$ mm the band is 145.6 MHz (4.94 % FBW), but the return loss is −1.75 dB. The pad-only launch used earlier reached −17.7 dB, but that launch was not the one fabricated. N = 2 has only one internal and one external coupling, which is not enough freedom to place two poles and also match them. The cascade has an extra degree of freedom at 1.4× the length, and it won the head-to-head comparison.

The fabricated design

The chosen geometry is $l_{\text{cav1}} = 33.939$, $l_{\text{cav2}} = 35.503$, $w_{\text{iris}} = 14.416$, $w_{\text{ext}} = 13.96$, $l_{\text{taper}} = 15.738$ and $w_{\text{ms}} = 1.6636$ mm, on a 165.78 × 45.79 mm board. Fig. 8 shows the fine-mesh result (2.5/0.45 mm volume mesh, 0.15 mm coax ports, 9.12 mm air, 61 points): fc 2.876 GHz, −3 dB bandwidth 182.3 MHz, worst in-band $|S_{11}|$ −11.15 dB, insertion loss 1.70 dB at the peak and 2.58 dB at the band edge. Table V (Sec. VIII) sets it against the specification, CST and the measurements.

Why it misses the specification. A lossy N = 3 coupling-matrix fit to the refined sweep of the untrimmed board (Fig. 9) puts the external coupling on target (+0.4 %), which exonerates the launch. It finds the internal coupling 4.8 % weak and the end cavities 17 MHz low. The

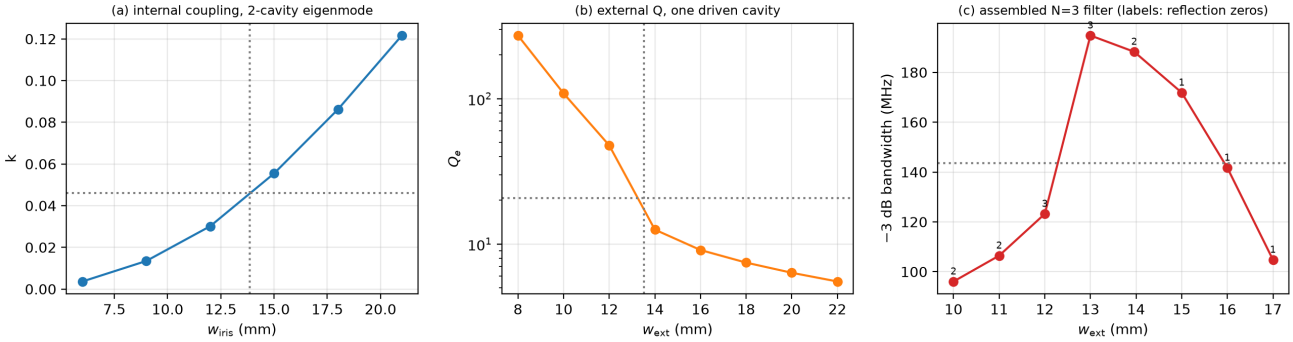


Fig. 3. (a), (b) The calibration observables are monotonic and invertible (dotted: synthesis target and inverted width). (c) The −3 dB bandwidth of the assembled filter, the quantity a black-box objective scores, jumps between 12 and 13 mm and then falls. Labels: number of reflection zeros. Coarse mesh.

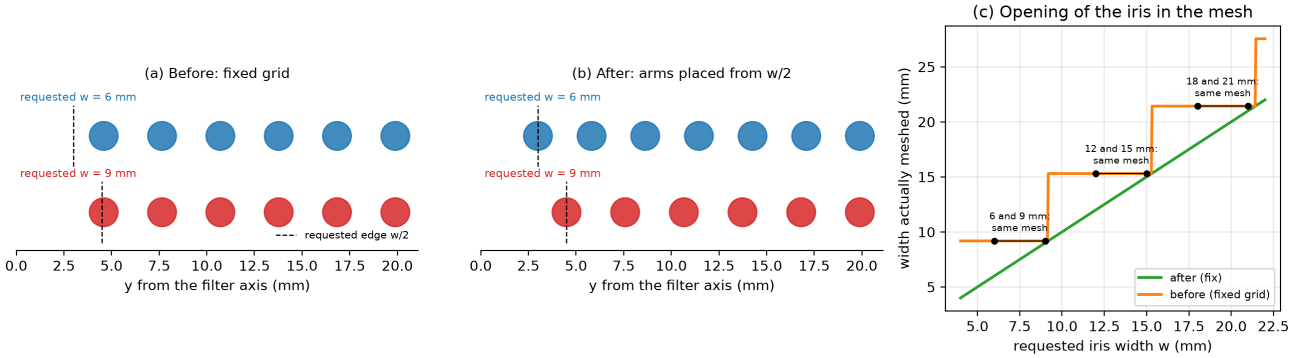


Fig. 4. Via-fence quantization. (a) With the fixed grid, requesting an iris of 6 mm or 9 mm placed exactly the same vias (dashed: requested edge, $w/2$). (b) With the fix the iris arms start at $w/2$ and every requested width gives a different mesh. (c) Opening actually meshed versus requested width: steps of about 6 mm before, a straight line after. Six requested widths fell on only three geometries.

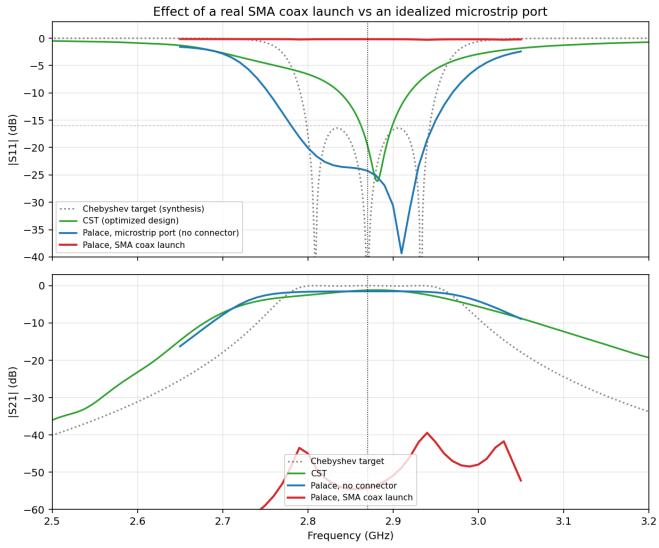


Fig. 5. First real-connector model. An ideal port hid a 41 Ω feed, and a cable-size pin larger than the board shorts the launch.

trim halves the detuning: the same fit on the fabricated geometry gives the end cavities 11 MHz low, the internal coupling 8 % weak and the external coupling 4.5 % weak (Q_e 22.6 against 20.6). The worst in-band point sits at the upper band edge, where the equiripple region ends near 2.935 GHz against a target edge of 2.942 GHz. Adding +0.6 mm of w_{iris} on top of the trim widened the band without helping. Adding +0.4 mm of w_{ext} on top of both was strongly harmful, and also pulled the end cavities down. No isolated w_{ext} trial was run. A coordinated local move remains open.

Why the insertion loss is high. The loss of a narrow-band filter is set by the unloaded Q, Eq. (5). With the dielectric and the copper in series,

$$\frac{1}{Q_u} = \frac{1}{Q_d} + \frac{1}{Q_c}, \qquad Q_d = \frac{1}{\tan\delta} = 270. \tag{6}$$

The Palace runs model the walls as perfect conductors, so their Q_u is capped by Q_d. The coupling-matrix fits returned 229–320, which brackets that cap; the 320 value exceeds it and reflects the fit's uncertainty, not lower loss. Real copper adds $Q_c \approx 540$ for a TE101 cavity 0.76 mm high with smooth copper [37], so $Q_u \approx 180$. Table III adds the terms up. The 0.5 dB we cannot attribute is the part of the Palace result that the matrix model does not have; we assign it to radiation from the open microstrip, taper and launch sections and did not separate it.

TABLE II
REPRESENTATIVE SILENT DEFECTS, HOW THEY WERE CAUGHT, AND THE GUARD ADDED

Class	Defect (effect)	Caught by	Guard now in code
Parameterization	Iris width quantized by grid filtering (search box ≈2 geometries); single iris width for $N \geq 4$ (spec unreachable); stale via defaults 0.8/1.5 mm	Scan showing identical meshes; coupling targets	Continuous via segments; per-iris parameters
Measurement / metric	Bandwidth taken first-to-last crossing, bridging two disjoint lobes; sweep window narrower than the band (fc reported = window centre, 48 MHz error charged nothing); truncated Q_e points giving an inversion outside its own bracket	Identical fc across evaluations; physics sanity	`n_lobes`, `band_truncated`, bracket check
Mode selection	Eigenmode pair = smallest gap picked a higher-order cluster: $k = 0.0028$ instead of 0.0583 (21×)	Inspecting the raw eigenvalue list	Window around f_0, then the two lowest modes
Mesh / boundary tagging	SMA end-cap classified by centroid swallowed the PTFE annulus at $r_{in} \geq 0.3773$ mm (S21 = −79.5 dB); faces left with no boundary condition (IL better than lossless); a domain with no material	Bisection over pin radius; IL > lossless	Face-extent test; unclassified-face report; domain-coverage guard
Model validity	Wave port on the trace self-normalizes, so W_{ms} was tuned to a 41 Ω line; surface-impedance BC applied to internal two-sided sheets (−7.8 dB IL on copper); 4.5–20 μm layers meshed as slivers	Real coax launch; Palace warning; one-factor ladder	Real SMA model; exterior-only conductivity; thin layers excluded
Plumbing	Adapter dropped soldermask/via-conductivity before the solver (two studies ran with the feature off); N=3 and N=4 results overwrote each other (shared tag); cached curve reused under a retuned label	Result unchanged by a real change	Order in every tag; geometry fingerprint on caches
Export	CST macro wrote the launch at 1000× (6 m coax, vias 30 m off-board), twice; taper dropped; coax without shield; PEC end caps on open walls	Cross-check of `.bas` vs mesh	34 file-level fab checks incl. magnitude test
Agent process	VBA properties guessed by name analogy; edits from a parallel agent absorbed unreviewed; comment overclaims; a process declared dead by a flawed `ps` filter	CST errors; adversarial self-review	Only properties seen in a real run; reviewed diffs
Convergence	Selection metric (S11 −13.37 dB) from a non-converged mesh; refined model gives −10.7 dB	CST/Palace audit	Fine-mesh final check

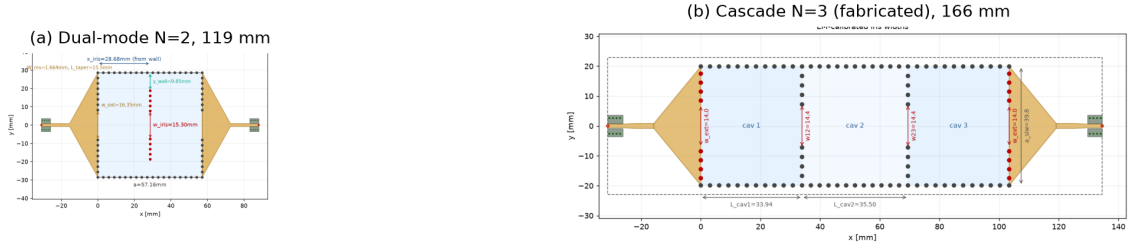


Fig. 6. (a) Dual-mode N=2: one square cavity whose TE_{102}/TE_{201} pair is split by a floating via row. (b) The fabricated N=3 cascade, with external apertures in red and the CPWG/SMA footprints at the board edges.

TABLE III
INSERTION-LOSS BUDGET AT 2.87 GHZ (DB).

Dielectric dissipation ($Q_u = 270$, 6.3 % bandwidth)	0.8
Mismatch at −11 dB	0.35
Launch and radiation (remainder of the Palace result)	≈0.5
Palace, perfect-conductor walls	**1.70**
Copper ($Q_c \approx 540$)	+0.4
Predicted, before the measurement	**2.1–2.3**
Measured (Sec. VIII)	**2.0–2.4**

The prediction was written before the board was measured. The 0.4 dB for copper is consistent with the 0.59 dB that an exterior-face copper model added. The result is close to the floor for this laminate, thickness and bandwidth, not a design error. Each loss mechanism points to an improvement:

- Qd is fixed by the laminate: $\tan\delta = 0.0037$ is high for a filter.
- Qc scales with the substrate height, and 0.76 mm is thin.
- The loss scales as 1/FBW, so a narrow band costs more.
- The band edges lose more (2.58 dB) because the group delay peaks there.

The 0.59 dB conductor-loss figure comes from a model that applies copper conductivity to the exterior faces only; the internal trace and ground sheets stay ideal, so it is a lower

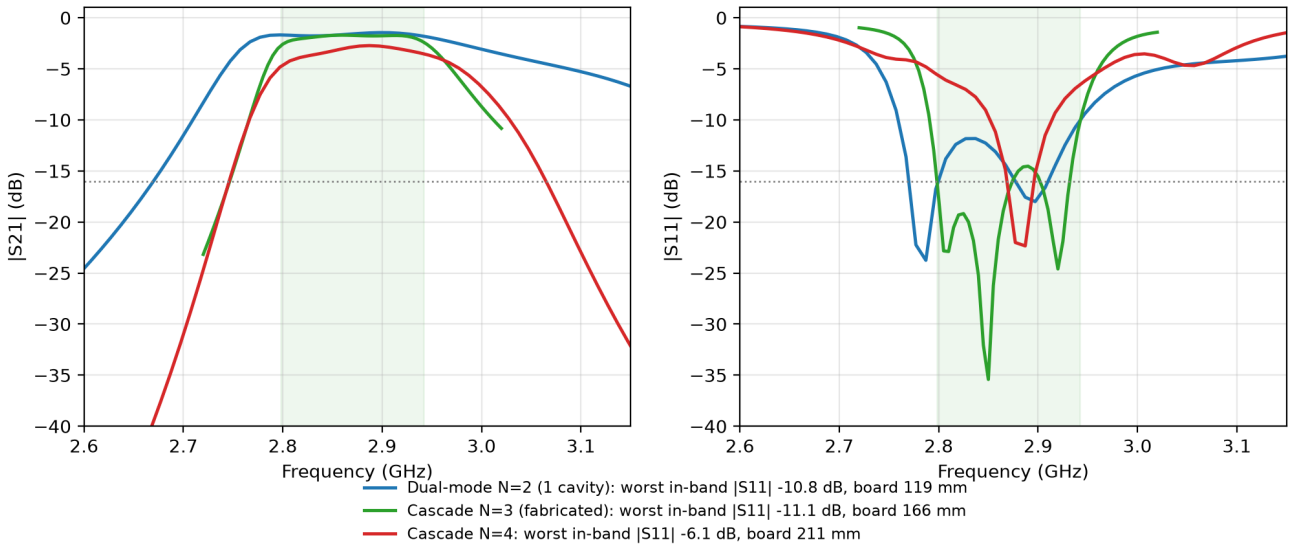


Fig. 7. Best designs of each topology under the same CPWG + SMA launch (Palace). The N=3 curve is the fine-mesh final. The other two are the production mesh.

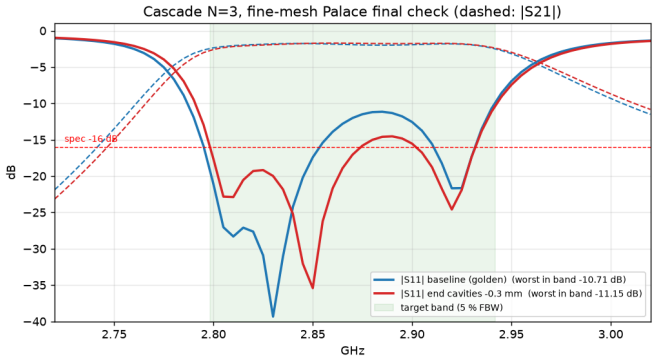


Fig. 8. Fine-mesh final check. The −0.3 mm end-cavity trim lifts the interior ripple from −11.2 to −14.5 dB. The worst point stays at the 2.94 GHz band edge.

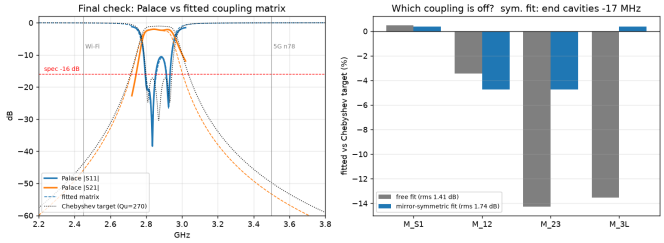


Fig. 9. Lossy coupling-matrix fit used to diagnose the return-loss gap: untrimmed board: external Q on target, k 4.8 % weak, end cavities 17 MHz low.

bound. A 1.52 mm RO4350B board only reaches IL ≈ 1.3 dB, because Qd still dominates. RT/duroid 5880 ($\tan\delta \approx 0.0009$) at 1.52 mm would bring it to about 0.5 dB.

Cross-solver check. CST Frequency Domain, run from the generated macro, agreed with Palace on fc (2.876 vs. 2.875 GHz) and IL (1.80 vs. 1.85 dB) on the untrimmed board. Getting there took ten debugging steps on the macro and solver setup, seven of them actual fixes, and a coax-only control model to separate port problems from filter problems. In Table V the CST run gives 2.878 GHz, 180 MHz and 1.80 dB against 2.876 GHz, 182 MHz and 1.70 dB for Palace on the fabricated geometry (CST is the untrimmed one). The time-domain solver was never made fully stable.

The connector is part of the design. Figs. 10 and 11 compare three launches on the same geometry: a bare microstrip feed, the CPWG launch without a connector, and the CPWG launch with the SMA. Without the connector only one reflection zero remains, and the worst in-band S11 is −6.3 dB (microstrip) or −5.1 dB (CPWG), against −11.1 dB with it. The filter was tuned with the connector's reactance in place. The bare CPWG port also shows a port-modelling trap: its first two Palace modes belong to the ground pours (S21 ≈ −78 dB), and the signal mode is mode 3.

Manufacturing. The same command-line tool writes the KiCad board, the gerbers, the drill files, a STEP solid, and 34 automated file-level checks [38]. The checks include drawn-versus-meshed geometry and CST-versus-mesh via positions. JLCPCB stocks RO4350B 0.76 mm but offers only 1 oz outer copper. That leaves the finished thickness at about 0.84 mm at the pads, against the SMA fork slot of 0.81 ± 0.1 mm, which is marginal and can be fixed by scraping 15–20 µm of copper under the connector legs. The no-mask requirement is carried by the gerbers and stated in the order notes. The boards were ordered from JLCPCB on Rogers RO4350B with an ENIG finish; the order cost about USD 120, laminate surcharge included. ENIG was chosen because the unmasked copper is exposed and ENIG is a durable barrier. We assumed that the plating, which sits on the air side of the copper, does not touch the cavity

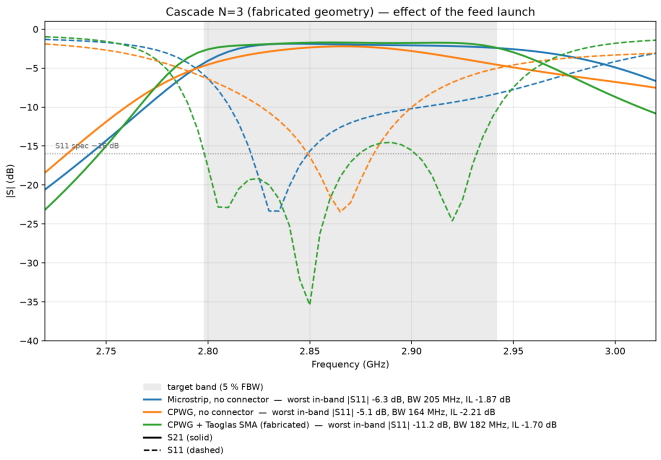


Fig. 10. Same geometry, three launches. Removing the SMA costs ≈5 dB of return loss because the design was tuned with it in place.

fields; the measurement cannot exclude it on the feed lines (Sec. VIII).

Measured results

Setup and calibration

The board (Fig. 12) was measured with a Keysight FieldFox N9917A handheld vector network analyzer (18 GHz), from 100 MHz to 5 GHz with 10 001 points, 10 kHz IF bandwidth and −15 dBm source power. The instrument was calibrated at the ends of the two test cables in full two-port SOLR (short-open-load-reciprocal) mode with a generic SMA kit. The short, open and load were connected on each port. Both cables end in SMA plugs and cannot mate with each other, so the thru was a female-female SMA adapter of unknown length and loss; SOLR is meant for exactly that (the R stands for a reciprocal, unknown thru). Isolation was omitted. The generic standards are not a characterised vendor kit; the instrument was set to the definitions of the 3.5 mm 85033D/E kit, the nearest it offers, so the true offset delays, open capacitance and load return loss are unknown. The cables end in SMA plugs that mate directly with the board's jacks, so the reference plane is the connector face, as in the simulations. Four states of the same board were saved: connectors soldered, or seated against the board without solder, each with the instrument calibration applied and without it. The raw open, short, match and thru were saved as well, at the same plane.

Software recalibration. The raw standards allow a second, independent correction. We solved the eight-term error model in software, with a one-port short-open-load calibration on each port and the unknown-thru step for the transmission terms [39], using scikit-rf [40], and applied it to the two uncalibrated boards.

How it works, in short. The analyzer does not see the board directly: between its receivers and each connector face sit cables, the switch and the instrument's own electronics. That path is modelled at each port as an *error box*, an unknown two-port (the eight-term model, seven independent terms per frequency). Calibrating means measuring known standards to solve for the two boxes; correcting means removing them from the filter measurement. Per frequency, in three steps:

1. *One port at a time (SOL).* At port 1 the measured reflection Γ_m and the true Γ are related by
$$\Gamma_m = e_{00} + \frac{e_{10}e_{01}\,\Gamma}{1 - e_{11}\Gamma},$$
with e_{00} the directivity, e_{11} the source match and $e_{10}e_{01}$ the reflection tracking. Multiplying out the denominator makes it linear, $\Gamma_m = e_{00} + \Gamma\,\Gamma_m\,e_{11} - \Gamma\,\Delta_e$ with $\Delta_e = e_{00}e_{11} - e_{10}e_{01}$, and the three standards of known Γ (short ≈ -1, open $\approx +1$, load ≈ 0) give a 3×3 system. Port 2 gives e_{33}, e_{22} and $e_{23}e_{32}$ the same way.
2. *Transmission from the unknown thru.* One term is left, the transmission tracking $e_{10}e_{32}$ that links the two boxes. Classic SOLT takes it from a zero-length thru, which does not exist here because the female-female adapter has a length and loss we do not know. The unknown-thru method [39] only uses that the adapter is reciprocal, $S_{21} = S_{12}$. With the boxes of step 1 known, that condition gives
$$(e_{10}e_{32})^2 = e_{10}e_{01}\;e_{23}e_{32}\;\frac{S_{21,m}}{S_{12,m}},$$
where $S_{21,m}$ and $S_{12,m}$ are the adapter's measured transmissions. The root has two solutions of opposite sign; at each frequency we keep the one that leaves the corrected adapter's phase closer to an ideal thru (within $\pm 90^\circ$). The sign only changes the phase of S_{21}, not the magnitude we report.
3. *Correction.* With both boxes known, each filter measurement is converted to transfer parameters T, in which a cascade is a product: $T_m = T_A\,T_{\mathrm{DUT}}\,T_B$, so $T_{\mathrm{DUT}} = T_A^{-1}\,T_m\,T_B^{-1}$, then back to S.

The instrument's switch terms were not measured, so the switch is assumed matched, as isolation is assumed negligible. The open and short follow the definitions the instrument was set to: a model of the 85033D/E (about 30 ps offset delay and 49 fF fringing capacitance; typical values, not read from the instrument). The result agrees with the instrument-corrected files to a mean $|\Delta\Gamma|$ of 0.03 for S_{11} and 0.04 for S_{22}, and to 0.17 dB for S_{21}, over 2.6–3.2 GHz (Fig. 13). Replacing the kit model by ideal $+1/-1/0$ standards changes the magnitudes by less than 0.03 dB (S_{21}) and 0.09 dB (S_{11}) and only moves the reference plane.

Comparison with TOSM. Classic TOSM (thru-open-short-match, Keysight's SOLT) uses the same standards and is identical in steps 1 and 3; only step 2 changes. TOSM declares the thru ideal and of zero length ($S_{11} = S_{22} =$

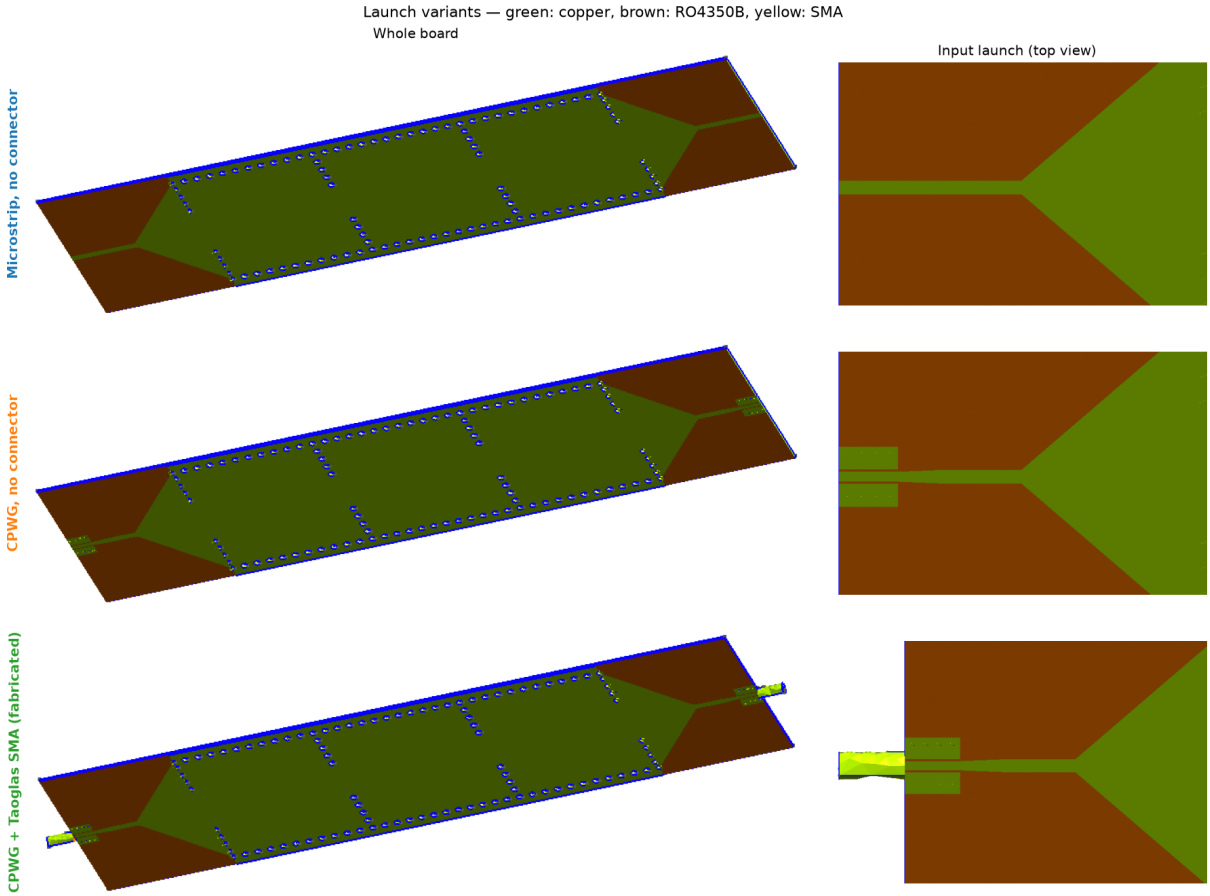


Fig. 11. Palace models of the three launches: whole board and input-launch close-up. Green: copper; brown: RO4350B; yellow: SMA connector.

0, $S_{21} = S_{12} = 1$), and with that assumption the thru measurement gives the transmission term directly:

$$S_{21,m} = \frac{e_{10}e_{32}}{1 - e_{11}\,e'_{22}} \;\Rightarrow\; e_{10}e_{32} = S_{21,m}\,(1 - e_{11}\,e'_{22}),$$

$$e'_{22} = \frac{S_{11,m} - e_{00}}{S_{11,m}\,e_{11} - \Delta_e},$$

where e'_{22} is the port-2 load match seen from port 1 through the thru, found by inverting the step-1 equation (12-term model; the reverse direction is analogous). With an unknown thru, by contrast, $S_{21,m}$ and $S_{12,m}$ share the same denominator, and their ratio cancels it together with the adapter's own transmission; this is why UOSM (SOLR) does not need to know the thru and TOSM does. If the thru is not ideal, TOSM loads its loss, phase and mismatch into the error terms.

To size that discrepancy we calibrated the same raw files both ways, with the same standard definitions (Table IV). TOSM changes the transmission term by +0.33 dB and −3.4° at 2.87 GHz and, by construction, leaves the corrected adapter at 0 dB and 0°; with the unknown thru the adapter reads +0.27 dB and −3.1°, and the instrument reads it at +0.18 dB and −2.2°. On the soldered board TOSM raises the insertion loss from 2.04 to 2.39 dB, narrows the band by 2.3 MHz and moves f_c by 2 MHz; the worst in-band S_{11} barely changes, and the best one deepens by 4 dB, because in TOSM e'_{22} comes from the thru's S_{11} and declaring the adapter's reflection zero (−22 dB after the unknown-thru correction) moves that reflection into e'_{22}. That the TOSM loss matches the instrument's (2.38 dB) is a coincidence: the instrument did not treat the adapter as ideal, as its +0.18 dB reading shows, and over the rest of the band TOSM is no closer to the instrument in transmission (mean ΔS_{21} +0.14 against +0.17 dB) and further in reflection (mean $|\Delta\Gamma|$ 0.041 against 0.030). The difference between the two methods, about 0.35 dB in insertion loss, is of the order of the 0.3 dB uncertainty assigned to S_{21} below.

Both corrections assume the same standard definitions, so their agreement checks the implementation and the saved raw files, not the standards. The offsets and the open's capacitance mostly move the reference plane, and the test above bounds their effect on the magnitudes. The adapter itself shows it: corrected with the 85033D/E definitions it has less than 1 ps of delay, and with ideal standards −64 ps; the reference plane moves by about 32 ps per port with the assumed definitions, while the magnitudes change by less than 0.09 dB. The load is different: a general-purpose SMA

TABLE IV
SOFTWARE CALIBRATION OF THE SAME RAW FILES WITH AN UNKNOWN THRU (THE REPORTED ONE) AND WITH TOSM, WHICH DECLARES THE FEMALE-FEMALE ADAPTER AN IDEAL ZERO-LENGTH THRU. SOLDERED BOARD; THE LAST COLUMN IS THE INSTRUMENT'S OWN CALIBRATION (SOLR, UNKNOWN THRU). THE ADAPTER ROW IS THE THRU ITSELF AFTER CORRECTION, AT 2.87 GHZ. MEAN DIFFERENCES OVER 2.6–3.2 GHZ.

	Unknown thru	**TOSM**	**Instrument**
$e_{10}e_{32}$ at 2.87 GHz	-12.04 dB $\angle 81.7°$	-11.71 dB $\angle 78.3°$	–
Corrected adapter S_{21}	$+0.27$ dB $\angle -3.1°$	0 dB $\angle 0°$ (by definition)	$+0.18$ dB $\angle -2.2°$
f_c (GHz)	2.8945	2.8926	2.8943
BW (MHz)	172.5	170.2	171.2
Peak IL (dB)	2.04	2.39	2.38
Best S_{11} (dB)	-18.8	-22.8	-16.8
Worst in-band S_{11} (dB)	-4.6	-4.5	-5.3
Mean ΔS_{21} vs. instrument (dB)	$+0.17$	$+0.14$	–
Mean $\lvert\Delta\Gamma\rvert$ S_{11} vs. instrument	0.030	0.041	–

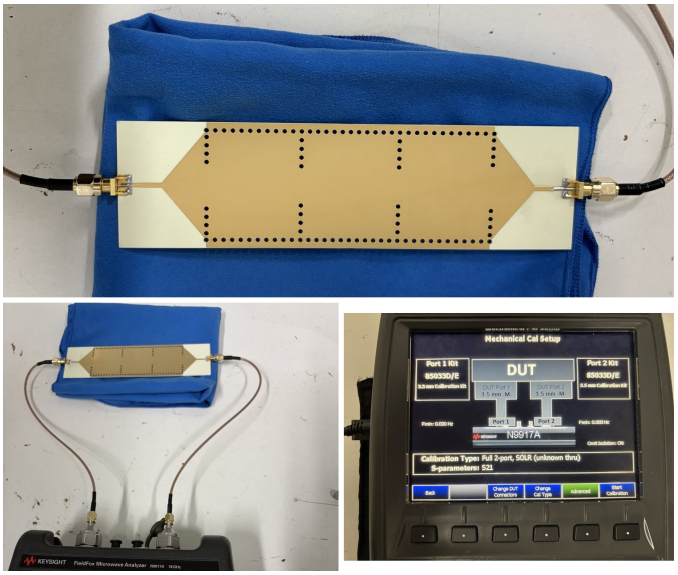


Fig. 12. Measurement setup. Top: the fabricated board with its connectors and test cables (no soldermask, ENIG finish). Bottom left: bench setup with the VNA. Bottom right: the instrument's calibration screen (3.5 mm 85033D/E definitions selected, full two-port SOLR with unknown thru).

load with a 25 dB return loss, which we did not measure, would leave a directivity error of 0.056, comparable to the dips we want to read. The check thru sets an empirical bound. After the instrument calibration it returns +0.18 dB in S_{21} and −18.4 dB in S_{11} at worst over 2.6–3.2 GHz, and the software calibration applied to the raw thru gives +0.26 dB and −17.6 dB. A passive thru cannot have gain, so both corrections are optimistic in S_{21}, and we take its uncertainty as about 0.3 dB. Reflections below about −18 dB are not resolved; the worst in-band S_{11} is good to about ±2 dB.

Comparison with simulation

Table V puts the specification, both simulators and the four corrected measurements through the same code (`siw measure`). Fig. 14 shows the curves.

How f_c and the bandwidth are read. One definition is used throughout the paper and for every source. On each curve we take the maximum of $|S_{21}|$ and draw a line 3 dB below it. The two crossings of that line on either side of the peak, linearly interpolated between sweep points, are the band edges f_1 and f_2. The bandwidth is $f_2 - f_1$, and the centre frequency is their midpoint, $f_c = (f_1+f_2)/2$, not the frequency of the maximum. With ripple in the passband the maximum sits on one of the poles and can be well off centre: in Palace the maximum is at 2.860 GHz and f_c = 2.876 GHz; on the soldered board the two coincide at 2.894 GHz. Only the crossings bounding the contiguous region around the peak count, so a band split in two is not reported as one wide band. Fig. 15 shows the construction on the curves.

Centre frequency and bandwidth. The measured band is centred at 2.894 GHz (soldered) and 2.898 GHz (unsoldered), against 2.876 GHz in Palace and 2.870 GHz in the specification. The +18 MHz (0.6 %) offset to the simulation corresponds to an effective permittivity about 1.3 % below the 3.66 used for design, well inside the spread between a laminate's design and process values; we did not test this. The −3 dB bandwidth is 171–174 MHz, 5–6 % below Palace.

Insertion loss. The peak S_{21} is −2.38 and −2.41 dB with the instrument calibration and −2.04 and −2.08 dB with the software one, so 2.0–2.4 dB within the 0.3 dB uncertainty above. Palace gives 1.70 dB. We predicted 2.1–2.3 dB before measuring (Sec. VII), so the measurement brackets the prediction. Section VII estimated the loss from the dielectric and a smooth-copper cavity. The forward model of the fitted coupling matrix can read the measured loss as an effective unloaded Q: keeping the Palace-fitted couplings and varying only Q_u, it gives 1.1 dB and three reflection zeros at Q_u = 270, 1.7 dB and two zeros at 180, 2.3 dB and one zero at 130 (Fig. 14b shows one). Read straight off the model, the measured 2.0–2.4 dB means Q_u of about 125–150. The matrix has no launch, and Palace shows 0.4 dB more than the matrix at Q_u = 229; removing that gives 150–185. The effective unloaded Q is therefore roughly 130–185, against 229 in the dielectric-only simulation and about 180 expected with smooth copper. We do not separate copper roughness, via-fence leakage, solder joints, the connector and the ENIG finish on the feed lines.

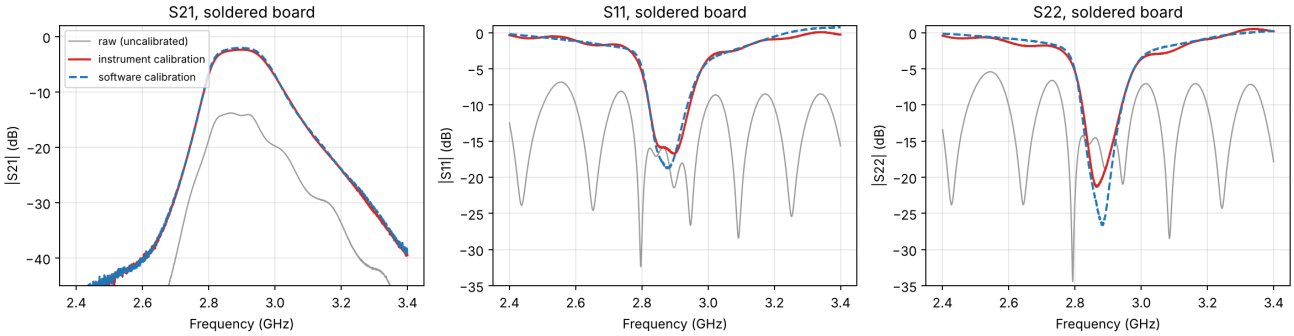


Fig. 13. Soldered board: raw measurement, instrument calibration and software calibration. The software result uses only the raw open, short, match and thru files.

TABLE V

FABRICATED CASCADE N=3 FILTER: SPECIFICATION, SIMULATION AND MEASUREMENT. EVERY COLUMN IS PROCESSED WITH THE SAME CODE: f_c IS THE MIDPOINT OF THE TWO CROSSINGS 3 DB BELOW THE PEAK S_{21}, AND THE BANDWIDTH THEIR DISTANCE. IN-BAND IS $2.870 \pm 1.4\%$ GHZ (THE SPECIFIED 5 % BAND). CST IS THE PRE-TRIM GEOMETRY. PALACE IN-BAND FIGURES ARE FOR THE FABRICATED GEOMETRY; ITS S_{21} AT 2.45/3.5 GHZ AND SPUR ARE FROM THE PRE-TRIM 0–8 GHZ SWEEP. THE SPUR IS THE LARGEST S_{21} BETWEEN 3.6 AND 5.0 GHZ. MEASURED COLUMNS: VNA = INSTRUMENT CALIBRATION, SW = SOFTWARE RECALIBRATION OF THE RAW FILES (SEC. VIII); S = CONNECTORS SOLDERED, U = UNSOLDERED, C = CALIBRATED. Q_L AND THE EFFECTIVE Q_u ARE DEFINED IN SEC. IX. THE SPECIFIED BANDWIDTH IS THE 0.1 DB RIPPLE BANDWIDTH; THE SPECIFICATION Q_L IS THAT OF THE IDEAL PROTOTYPE AT −3 DB (199 MHZ), NOT 1/FBW = 20.

	Spec.	**CST**	**Palace**	**VNA SC**	**VNA UC**	**SW S**	**SW U**
Centre frequency (GHz)	2.870	2.878	2.876	2.894	2.898	2.895	2.898
−3 dB bandwidth (MHz)	143.5	180	182	171	172	172	174
Fractional bandwidth (%)	5.0	6.2	6.3	5.9	5.9	6.0	6.0
Loaded $Q_L = f_c/\mathrm{BW}$	14.4	16.0	15.8	16.9	16.9	16.8	16.7
Effective Q_u (from IL)	–	173	179	138	136	159	155
Peak S_{21} (dB)	–	−1.80	−1.70	−2.38	−2.41	−2.04	−2.08
Worst in-band S_{21} (dB)	–	−3.76	−2.58	−7.14	−7.62	−6.79	−7.26
Worst in-band S_{11} (dB)	≤ -16	−6.9	−11.1	−5.3	−4.8	−4.6	−4.1
Worst in-band S_{22} (dB)	–	−6.9	–	−5.2	−4.5	−5.0	−4.3
Best S_{11} (dB @ GHz)	–	−17.3 @ 2.877	−35.4 @ 2.850	−16.8 @ 2.899	−17.1 @ 2.884	−18.8 @ 2.881	−20.1 @ 2.873
S_{11} zeros < -10 dB	–	1	3	1	1	1	1
S_{21} at 2.45 GHz (dB)	–	−60.7	−60.9	−45.6	−45.8	−44.9	−45.0
S_{21} at 3.5 GHz (dB)	–	−39.3	−39.5	−43.3	−43.5	−41.6	−42.3
Strongest spur (dB @ GHz)	–	−2.5 @ 4.63	−3.8 @ 4.61	−4.4 @ 4.68	−4.2 @ 4.68	−4.5 @ 4.68	−4.4 @ 4.68

Return loss. The trace shows one broad minimum with a shoulder, −17 dB at best with the instrument calibration (−19 to −20 dB with the software one), and the worst in-band S_{11} is −5 dB. The three simulated zeros are at −23 to −35 dB, below the −18 dB floor of the calibration check, so the measurement cannot confirm them one by one; the matrix model says that this much loss merges them into one, which is what the trace looks like. The 24 MHz offset from the specified band adds to the in-band figure. The ports differ slightly (best S_{22} −21 dB against −17 dB for S_{11} on the soldered board), within the calibration floor.

Soldering. Soldering the connectors changes the calibrated result by 3 MHz in centre frequency, 0.03 dB in insertion loss and 0.5 dB in worst in-band S_{11}. At 2.9 GHz the contact of the seated connector is adequate, and the fork-slot fit discussed in Sec. VII did not become a problem.

Out of band. The first spurious passband appears at 4.68 GHz and −4.4 dB, against 4.61 GHz and −3.8 dB in Palace and 4.63 GHz and −2.5 dB in CST. Above the passband the measurement stays within a few dB of Palace (−45.6 against −44.2 dB at 3.7 GHz, −41.2 against −41.7 dB at 4.0 GHz, −43.0 against −39.5 dB at 3.5 GHz). Below it the simulated rejection keeps falling (−61 dB at 2.45 GHz, −92 dB at 2.0 GHz), while the measured one levels off at −44 to −48 dB between 2.2 and 2.5 GHz and is −55 dB at 2.0 GHz, well above the instrument floor (below −70 dB). This points to coupling around the filter (open feed regions, cables) or leakage through the via fence, which the models do not contain.

Effect of the launch

Table VI adds the measured board to the launch simulations of Sec. VII.

The simulations without a connector use ports that normalise to the planar line and are not comparable in absolute terms with a measured board, which has a connector and a real 50 Ω reference. They show that the connector's reactance is part of the tuning. The measurement shows that the board with its connector agrees with the connector simulation in centre frequency, bandwidth and spurious bands, and not in return loss.

All figures in this section are regenerated from the files in `measurements/2026-10-14_vna/` by `siw measure --paper`.

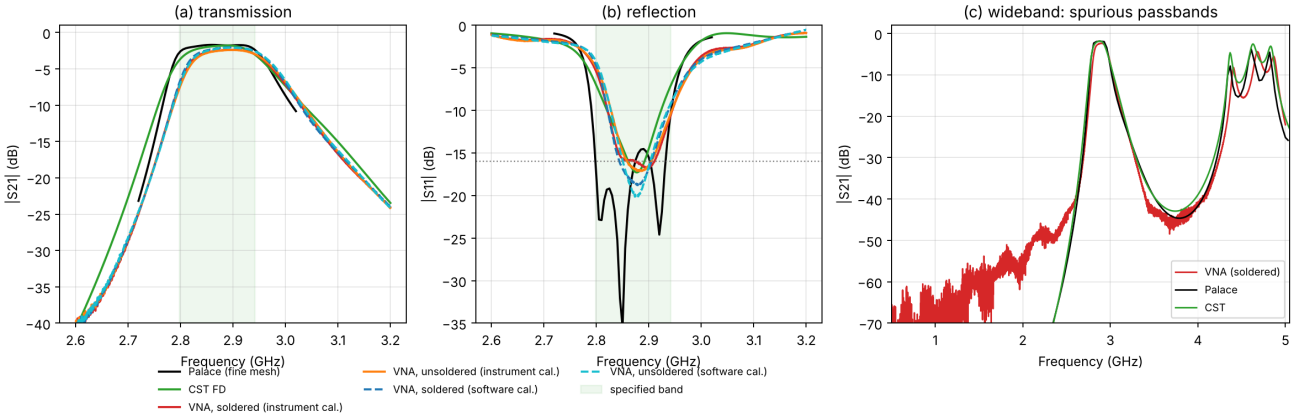


Fig. 14. Measured and simulated response. (a, b) The fabricated board (instrument and software calibration, soldered and unsoldered) against Palace and CST. (c) Wideband transmission: the spurious passbands at 4.4–4.9 GHz appear in the measurement at the simulated frequencies.

TABLE VI
EFFECT OF THE LAUNCH ON THE SAME FILTER GEOMETRY. THE FIRST THREE ROWS ARE PALACE SIMULATIONS (THE PORTS OF THE TWO CONNECTOR-LESS VARIANTS SIT ON THE PLANAR LINE AND NORMALISE TO ITS OWN IMPEDANCE, SO THEY SHOW THE FILTER BEHIND AN IDEALLY MATCHED FEED). THE LAST TWO ROWS ARE THE FABRICATED BOARD (INSTRUMENT CALIBRATION). S_{21} IS THE PEAK AND S_{11} THE WORST VALUE IN THE SPECIFIED BAND.

Launch	f_c (GHz)	**BW (MHz)**	S_{21} (dB)	S_{11} (dB)
Microstrip, no conn.	2.897	205	−1.87	−6.3
CPWG, no conn.	2.875	164	−2.21	−5.1
CPWG + SMA, sim.	2.876	182	−1.70	−11.1
CPWG + SMA, sold.	2.894	171	−2.38	−5.3
CPWG + SMA, unsold.	2.898	172	−2.41	−4.8

QUALITY FACTOR: LOADED AND UNLOADED Q

Two quality factors summarise the response. The **loaded Q** of the filter, $Q_L = f_c/\mathrm{BW}_{-3\,\mathrm{dB}}$, says how narrow the passband is with the ports attached. We read it straight off $|S_{21}|$: the dotted line in Fig. 15 sits 3 dB below each curve's peak, and its ends are the band edges. The **unloaded Q** Q_u measures each resonator's own loss and sets the insertion loss (Eq. 5). The two Q rows of Table V and Fig. 16 come from the same code (`siw measure`) for all four sources.

In the full filter the two play different roles:

- Q_L is set by the couplings (k and Q_e). It defines the bandwidth and barely depends on Q_u as long as $Q_u \gg Q_L$.
- Q_u does not change the bandwidth, but it sets the insertion loss. Writing Eq. 5 in terms of the loaded Q of the ripple band, $Q_{L,\mathrm{ripple}} = 1/\mathrm{FBW}$,

$$\mathrm{IL}_0 \approx 4.343 \sum_i g_i \frac{Q_{L,\mathrm{ripple}}}{Q_u}\ \mathrm{dB}.$$

  The narrower the filter (higher Q_L) or the lossier each resonator (lower Q_u), the higher the loss.

Theoretical loaded Q. The specified 5 % is the 0.1 dB *ripple* bandwidth, 143.5 MHz, whose Q is 1/FBW = 20. At −3 dB the ideal third-order Chebyshev prototype is $\cosh(\frac{1}{N}\operatorname{arccosh}\sqrt{(10^{0.3}-1)/(10^{0.01}-1)}) = 1.39$ times wider: 199 MHz, or $Q_L = 14.4$. That is the value to compare with, because every bandwidth in this paper is a −3 dB one. Palace gives 15.8, CST 16.0 and the board 16.9 (16.8 with the software calibration). Against that reference the −3 dB band is not wider than designed but 9 % (Palace) to 14 % (board) narrower, consistent with the internal coupling the matrix fit finds slightly below target ($k_{12} = 0.042$ in Palace and 0.044 on the board, against 0.046). Earlier comparisons against 143.5 MHz mix the two bandwidth definitions.

Effective Q_u. We solve Eq. 5 for Q_u with the measured peak loss. The FBW in that equation is the ripple one, so the measured −3 dB FBW is divided by 1.39. On the ideal lossy prototype the procedure returns 270 for $Q_u = 270$ and 181 for 180. Because all loss (mismatch, launch, radiation, connector) is charged to the resonators, the result is an *effective* Q_u, a lower bound on the resonator's own. Palace gives 179 and CST 173. Both simulations have perfectly conducting walls, whose ceiling is the dielectric, $Q_d = 270$; the gap is the mismatch and launch terms of the loss budget (Table III). The board gives 138 with the instrument calibration and 159 with the software one, inside the 130–185 range the coupling-matrix forward model gave in Sec. VIII. The board sits at 0.77–0.89 of Palace; that is what copper, the ENIG finish and the real connector add, none of which the simulations contain. The smooth-copper reference, $1/Q_u = 1/Q_d + 1/Q_c$ with $Q_c \approx 540$, gives 180 for the resonator alone.

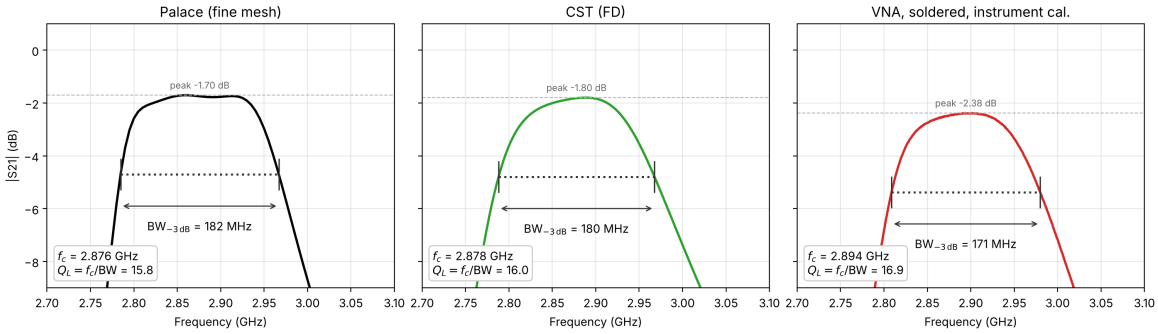


Fig. 15. Loaded Q read from $|S_{21}|$. The dotted line is 3 dB below each curve's peak and runs edge to edge; the arrow marks the bandwidth and the box gives $Q_L = f_c/\mathrm{BW}$.

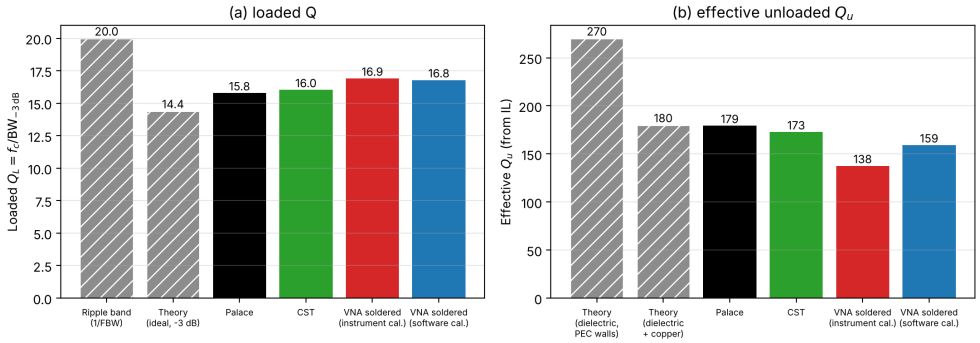


Fig. 16. Quality factors of the four sources against theory (hatched bars). (a) Loaded Q: the comparable value is the ideal prototype at −3 dB (14.4), not 1/FBW (20, ripple band). (b) Effective Q_u solved from the insertion loss, against the dielectric limit (270) and dielectric plus smooth copper (180).

Discussion: gaps and room to improve

What was AI-driven, and what was not. The agent wrote about 11.5 k lines of design code, launched and monitored the Palace eigenmode and driven solves, read solver logs, localized most of the defects in Table II, and produced the fabrication files. The engineer chose the specification, topology, laminate and vias, decided when to stop tuning, ran every CST check, and more than once rejected a confident but wrong explanation. The design physics that finally worked is classical. The acceleration came from automation and diagnosis, not from new electromagnetics. Stating this plainly is more useful than overclaiming.

Agent failure modes observed.

- Confident misdiagnoses: two wrong explanations of one stack failure were each falsified by a single re-run.
- APIs guessed by analogy: VBA properties that do not exist.
- Documentation drift: headline numbers kept in prose after the code had changed. The repository's status files had to be rewritten several times.
- Unreviewed edits from a parallel agent absorbed into a commit.
- Premature optimism about untested geometries.

Each was caught by a physics check, a solver error or a human question, which argues for keeping those checks in the loop.

Gaps.

- One board and one measurement session: no spread across boards, and the S21 scale is uncertain by about 0.3 dB (Sec. VIII).
- The calibration standards were a generic SMA short, open and load, not a characterised kit. The calibrated thru returns −18 dB, so reflection zeros deeper than that are not resolved, and the worst in-band return loss is good to about ±2 dB.
- The specification is not met: the board measures −5 dB worst in-band return loss against −16 dB (simulated: −11.1 dB), and 171 MHz bandwidth against 143.5 MHz.
- S11 varied by about 5 dB across mesh and port models, while fc, bandwidth and IL moved by less than 10 MHz and 0.3 dB. The measurement then differed from Palace by +18 MHz in fc, 0.3–0.7 dB in IL and 6 dB in S11, so the model uncertainty was not conservative.
- Qe is calibrated with a dielectric-only Qu and is slightly over-estimated.
- Conductor loss is modelled only on exterior faces. Thin fabrication layers (mask, ENIG) could not be resolved.
- Spurious passbands start at 4.4 GHz (−4 dB, confirmed by the measurement) and need a low-pass filter in a real system.

- The measured loss is not apportioned between copper roughness, via-fence leakage, solder joints, connector and ENIG on the feed lines; the low-side rejection floor of about −46 dB (simulated: −61 dB) points to coupling around the filter that no model contains.
- There is no controlled comparison against an engineer working without the agent, and no accounting of tokens or cost.
- This is a single case study: one engineer, one model family, one filter topology. It shows what can happen, not how often.
- The taxonomy is self-reported by the same engineer–agent pair that made the mistakes. Git corroborates that each fix happened, but not the classification. It also lists only caught defects; the physics checks may still miss a comparable class of error.
- The code has no regression test suite. Its safety net is the physics sanity checks and the automated fabrication checks.

Improvements.

- A coordinated local correction of w_{ext}, l_{cav1} and w_{iris}, driven by the coupling-matrix fit, plus re-centring the band: the +24 MHz offset from 2.870 GHz is of the order of 0.5–1 mm of cavity length (27 MHz/mm measured at the end cavities, about 46 MHz/mm for the ideal cavity).
- Fitting the coupling matrix to the measurement needs a model of the launch. With the plain matrix the fit is ill-conditioned (Q_u between 57 and 85 for two nearly identical curves), so we read the loss through the forward model.
- Adaptive frequency sweeps and y-symmetric half-models, for roughly 2× memory and 5–10× fewer solves, which would bring the cascade within GPU limits.
- A GPU with more memory.
- A lower-loss or thicker laminate.
- Space-mapping or surrogate models [2] trained on the calibration data.
- Repeating the measurement with a characterised SMA calibration kit, or with on-board thru-reflect-line standards, to resolve the zeros.
- Making the physics sanity checks a mandatory step of the agent loop rather than a habit.

Conclusion

An LLM coding agent took an SIW bandpass filter from specification to fabrication files, with one engineer supervising, and the board was then manufactured and measured. Two things decided the design outcome. The first was replacing black-box search over a discontinuous objective with calibration of smooth observables on small structures. The second was catching a long series of silent defects with physics checks. The measurement confirms the simulations where the physics is simple: the spurious bands fall at the predicted frequencies and the centre frequency is within 1 % of the specification. It also shows what the simulations did not capture: 0.3 to 0.7 dB more loss than Palace, which in the matrix model merges the three reflection zeros into one, so the worst in-band return loss is 5 dB, not the simulated 11 dB and not the specified 16 dB. The generic calibration standards limit the reading of the zeros themselves (floor about −18 dB), not the 5 dB. The independent software recalibration agrees with the instrument's on average within 0.2 dB in S_{21} (up to 1 dB on the steep band edges), so this conclusion does not rest on one calibration. The code, data and scripts that produced every number are in the repository, and `siw measure` reproduces the measured columns of Table V.

Acknowledgment

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